

Hi! I'm Degly

Extended Realities Enthusiast
and Systems and Computer Engineer



I am
an
innovator

Check out my projects



Degly Pava



[Download my CV](#)

<https://www.deglypava.com>

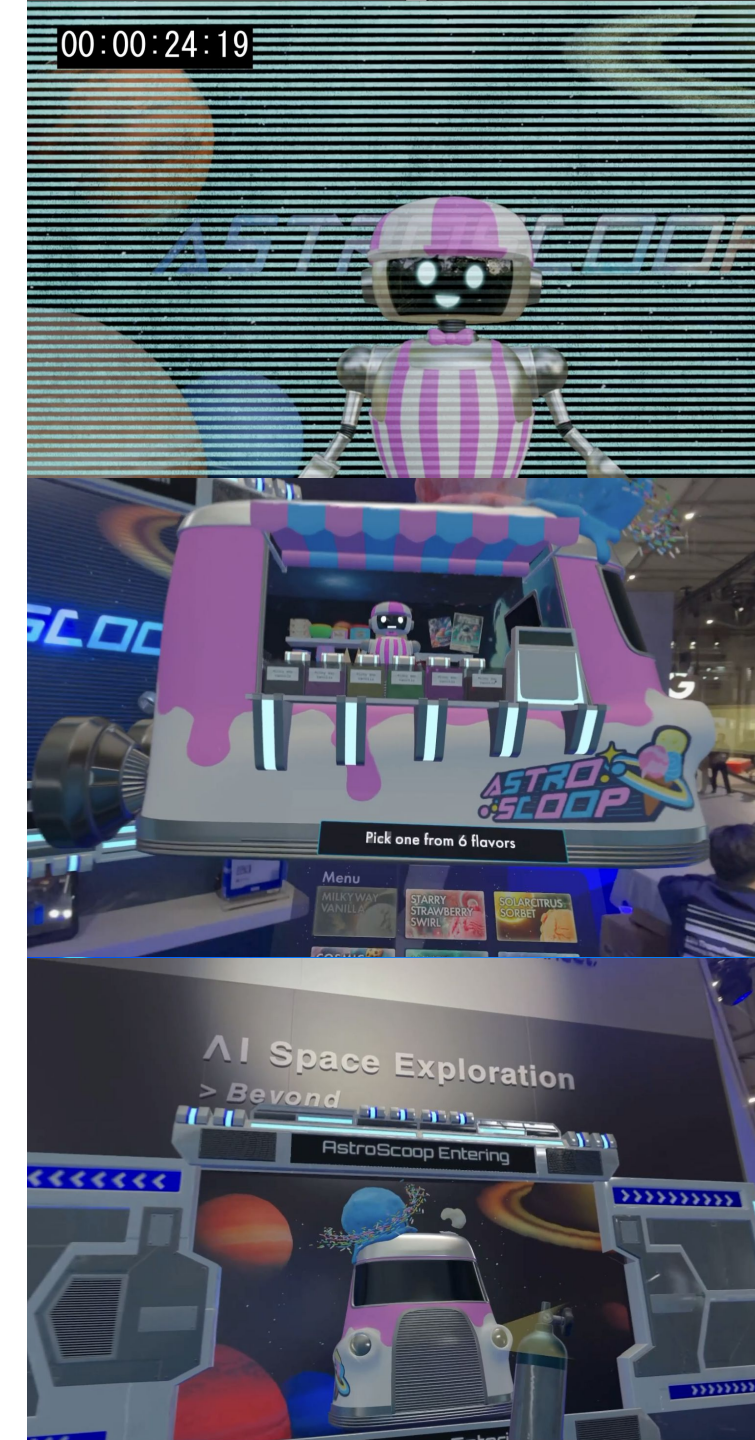
Check my website for an interactive portfolio

- Created a personal website to showcase projects and skills, featuring a custom-generated **GLSL + Three.js** animated background.
- Adapted and animated a **Samsung AR Emoji** as the centerpiece avatar — fully rigged and dancing with **Mixamo** animations.
- Modeled a stylized 3D version of my name in **Blender**, viewable in the real world using **Google's AR Model Viewer**.
- Boosted engagement with **interactive elements** and real-time animations using **Three.js**, creating a dynamic and immersive web experience.



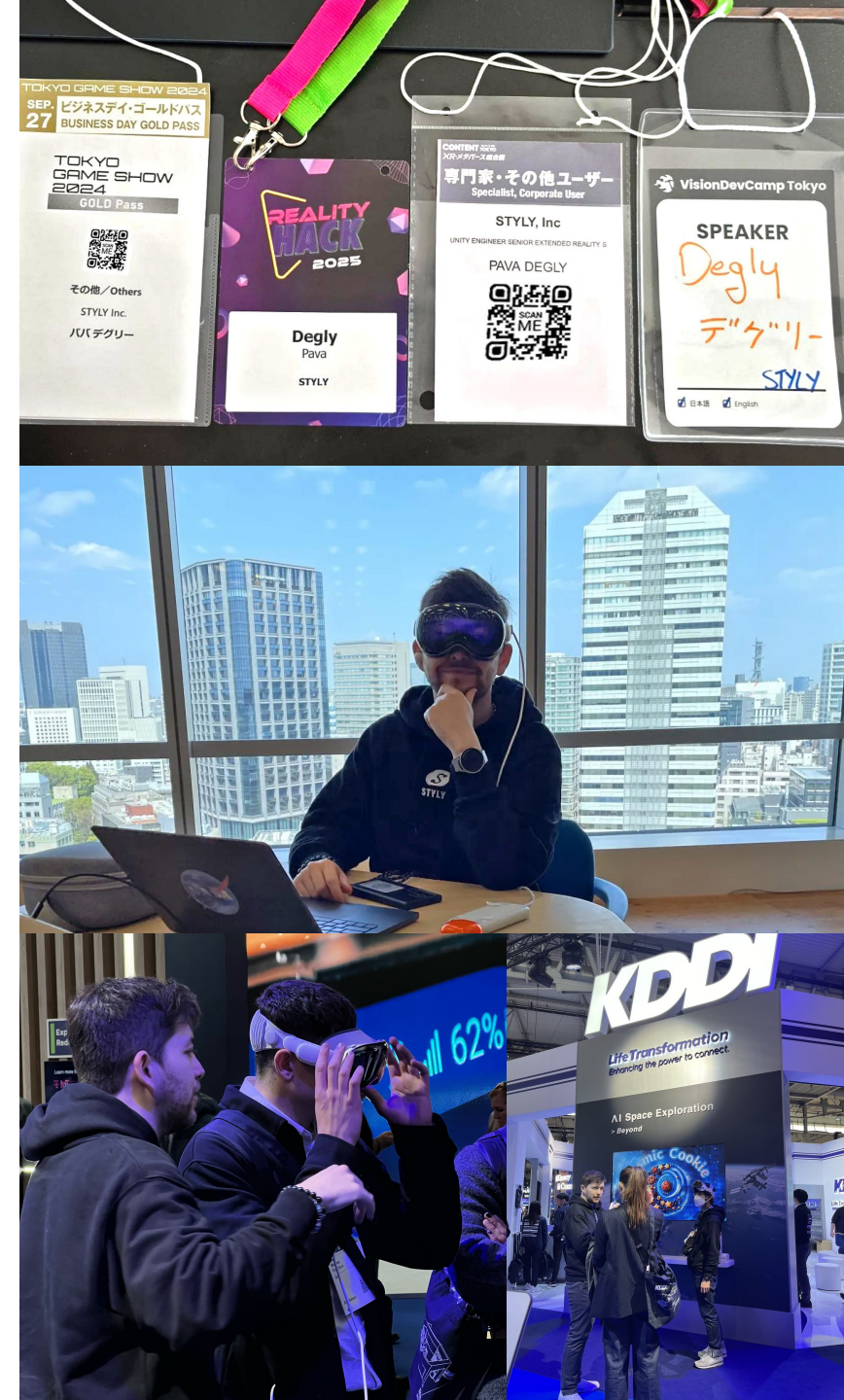
Astroscoop – Apple Vision Pro Game Demo

- Created a space exploration **game demo** for **Apple Vision Pro**, built in **Unity** and showcased live at **MWC 2025** as part of KDDI's XR initiative.
- Development was done in STYLY, gameplay and immersive interactions for mixed reality, blending real-world environment with spatial storytelling.
- Personally demoed the experience on the show floor, strengthening **STYLY's global presence** in immersive gaming and developer ecosystems.



Global XR & Game Dev Events

- Represented teams and projects at major events including **Tokyo Game Show, MIT Reality Hack, MWC, SushiTech**, and more.
- Mentored over **700 developers** across international hackathons and conferences, supporting award-winning projects and building community connections.
- Built strong industry ties with **Oculus Start developers**, global XR creators, and academic partners from **MIT, Harvard, and Parsons** — helping expand XR adoption worldwide.



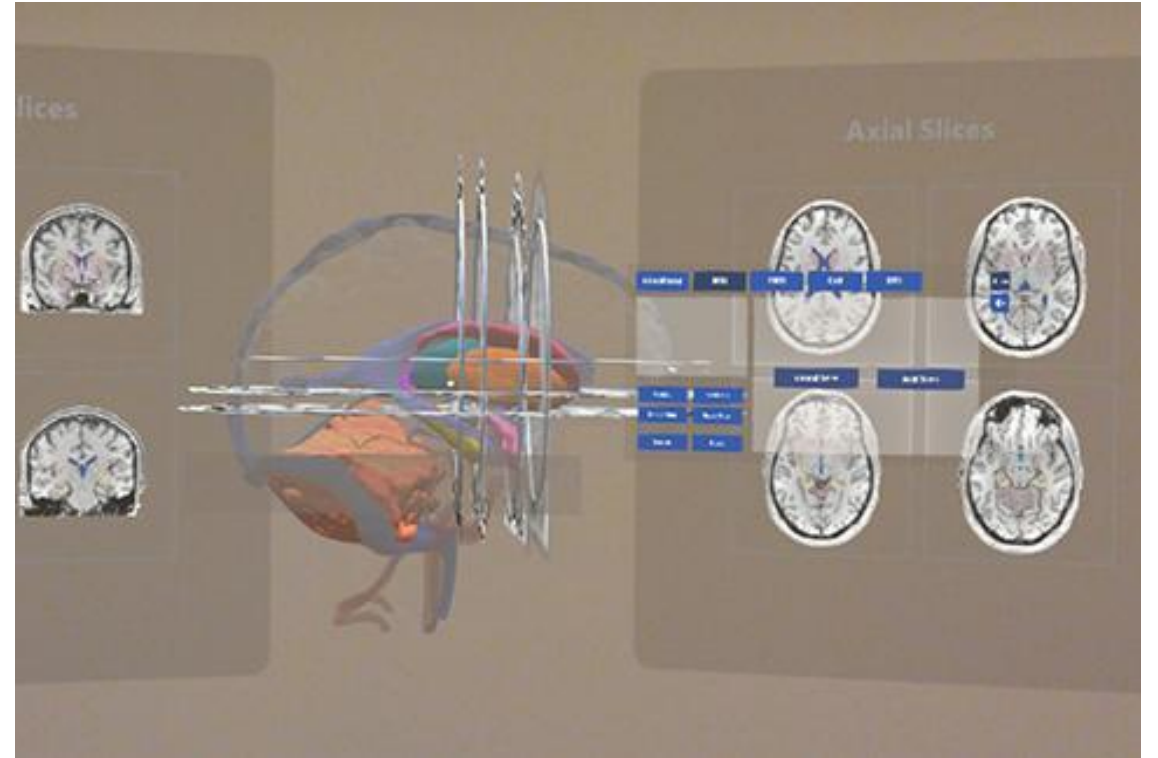
Unity AR and Game Development

- Built core AR systems for mobile games like *AR謎解きゲーム*, *AnimalPal*, and *ARgamemadeEZ*, released in **173 countries** and other featured at **Universal Studios Japan**.
- Collaborated with major studios including **CAPCOM** and **Nintendo**, working directly with designers and external developers.
- Innovated on AR glasses tech via **Qualcomm's Pathfinder Program**, focusing on **multi-device collaboration**, seamless UX, and mobile-to-headset connectivity.



Monster Hunter World:
Icerborne AR

HoloBrain 2.0



MRI-Derived Brain Visualization on HoloLens

- Transformed **MRI brain scans** into immersive **3D volumetric animations** using **C++ shaders** and **Blender**.
- Applied **3D math**, **OpenGL**, **Unity**, and **linear algebra** to develop a **holographic educational tool** for HoloLens.
- Designed an intuitive mixed reality experience, enabling users to explore a floating brain model in real space for **interactive medical learning**.



HoloLens AR Car Tracking Game

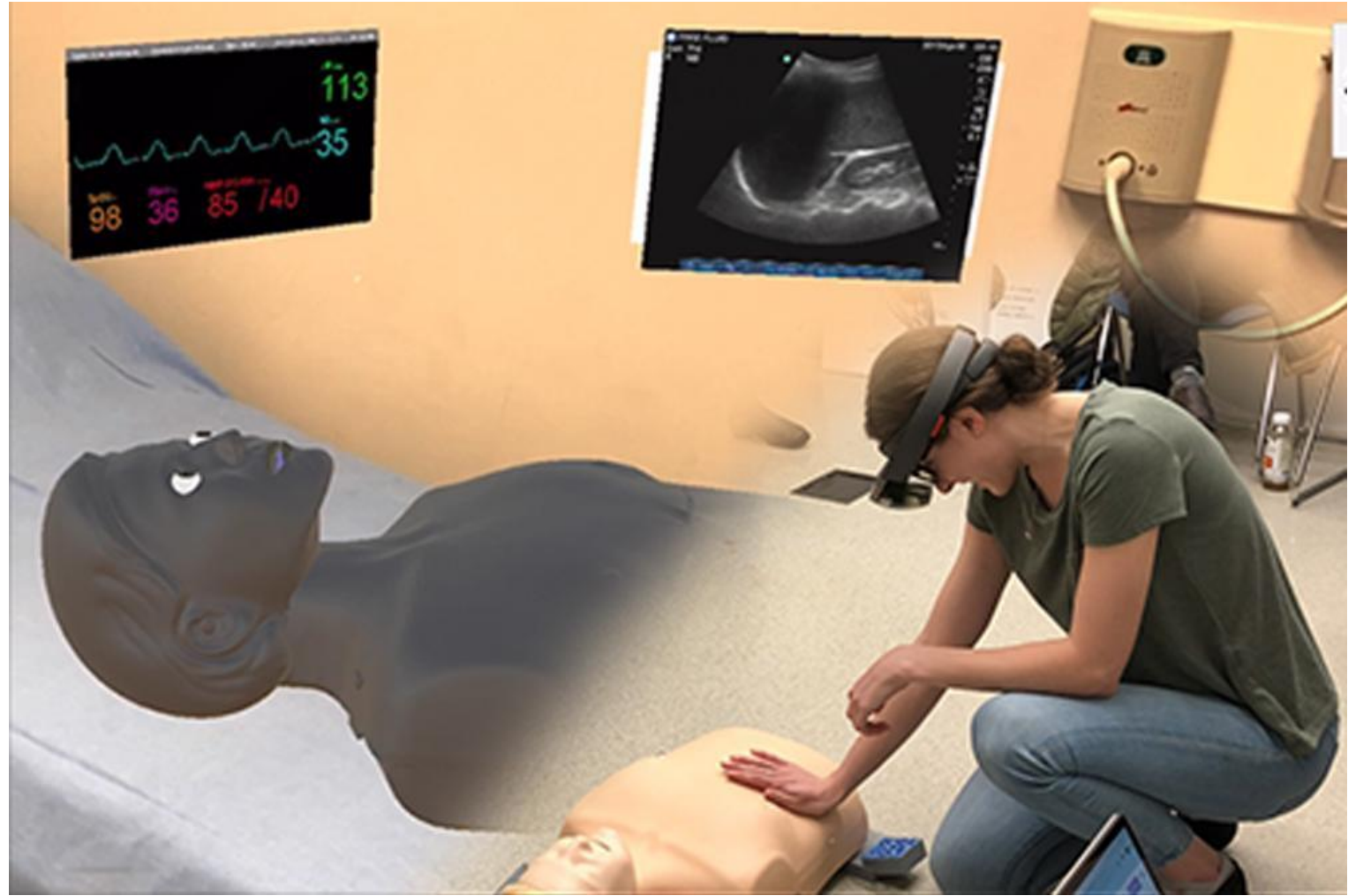


HoloLens AR Car Tracking Game

- Built a **mixed reality racing game** on HoloLens using **Vuforia Image Trackers** and **OpenXR**.
- Integrated a custom **Arduino-powered car**, enabling real-world tracking and dynamic AR interactions.
- Designed *Mario Kart*-inspired mechanics where players collect power-ups via physical AR markers.
- Showcases skills in **AR development**, **hardware integration**, and **creative XR game design**.

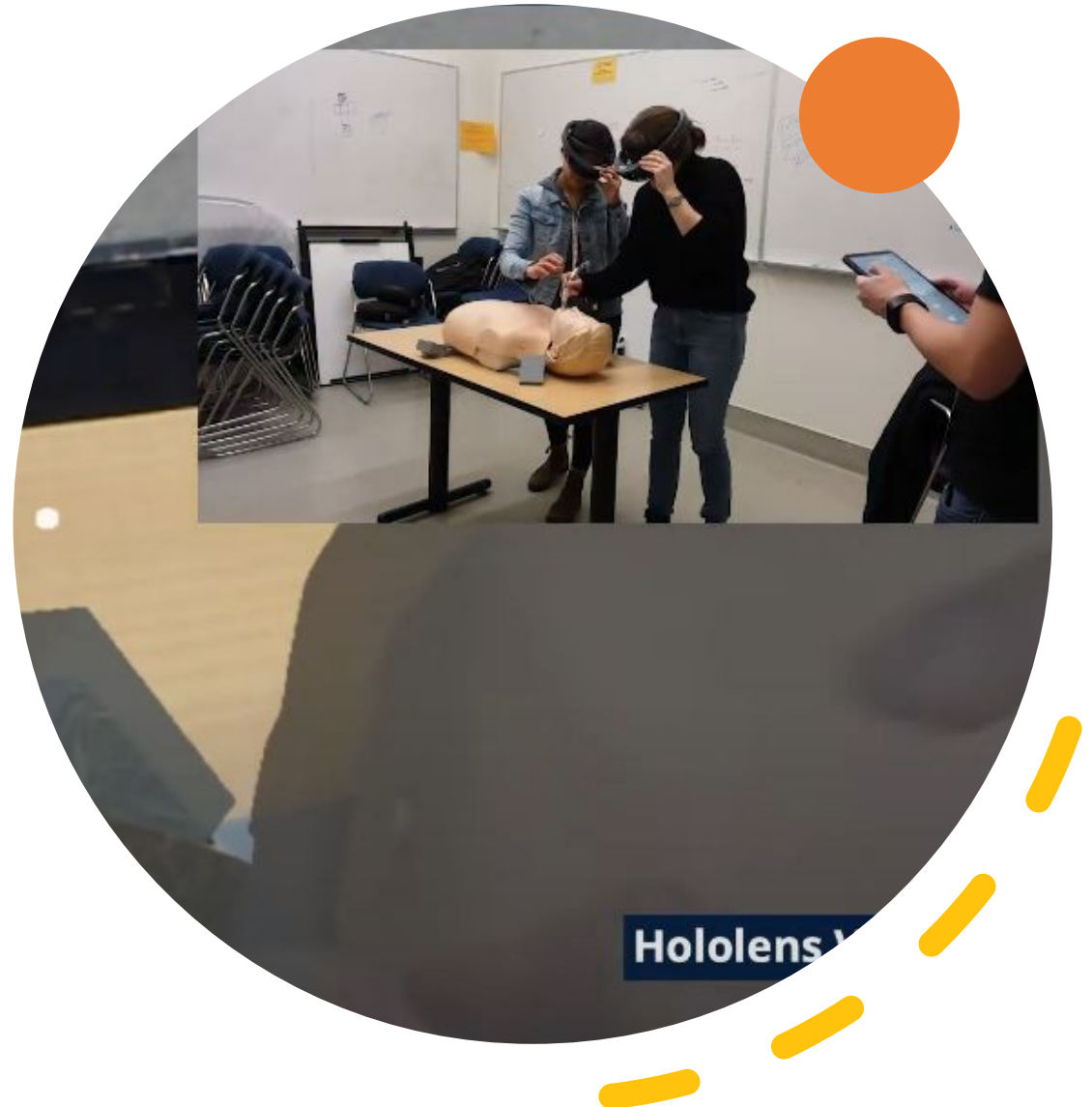


Mr Manikin Online Clinical Simulation



Mr. Manikin – Online Clinical Simulation

- Developed an AR-based mobile simulation featuring “Annie,” a virtual mannequin instructor for teaching medical techniques like PCR.
- Built using **ARCore + ARKit** for cross-platform support (Android/iOS) and integrated **Photon** for real-time **multiplayer collaboration**.
- Worked in a distributed team of 10+ using **Slack, Jira, and Git**, applying agile workflows during the game design and development phases.

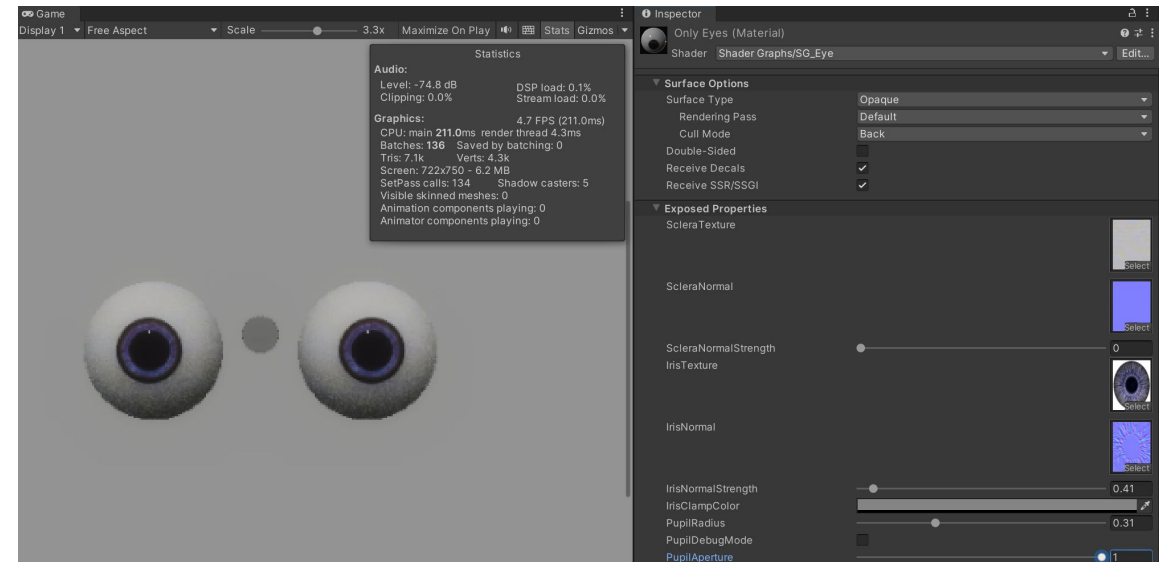


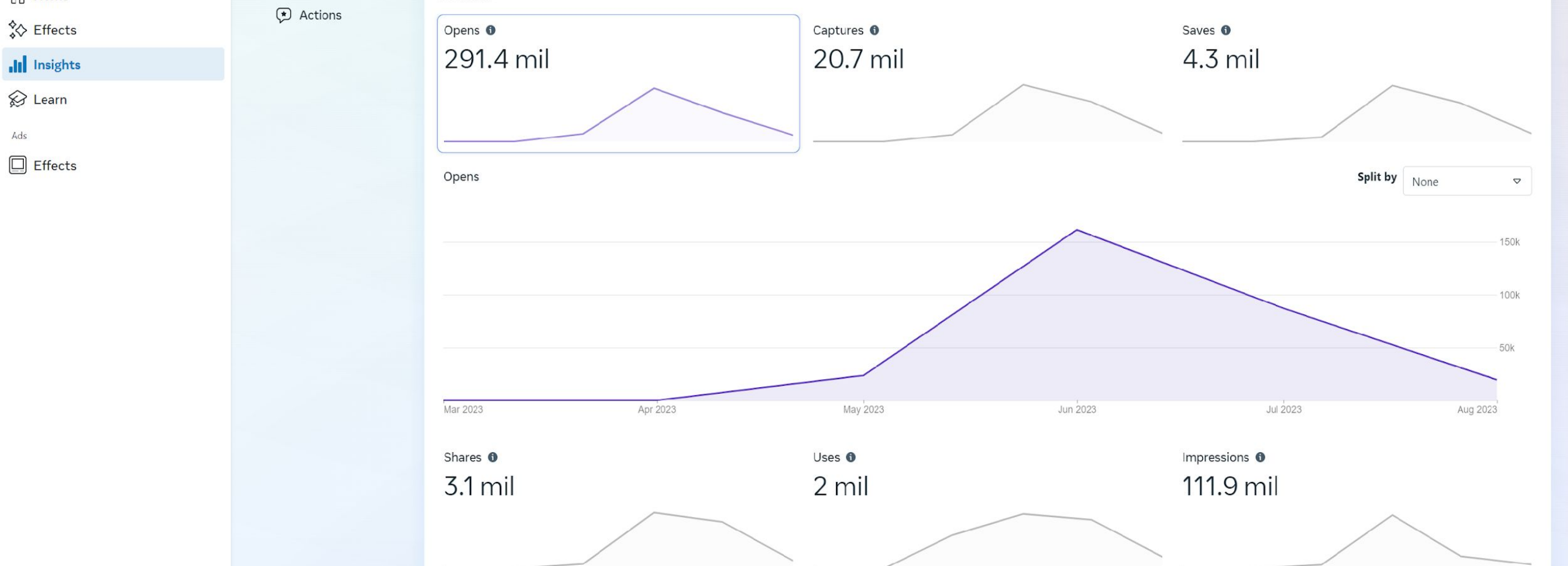


Pupil Mimicry in HCI with Virtual Agents

Pupil Mimicry in HCI with Virtual Agents

- Cited in academic work and research; presents opportunities for **emotional AI** and **character empathy in games**.
- Built a real-time Unity environment using **OpenXR** and **Blender** to study **pupil mimicry** with virtual agents.
- Integrated **eye-tracking** via C++ and C# for lifelike pupil dilation and behavioral responsiveness.
- Demonstrated that users **unconsciously mirror pupil changes**, enhancing emotional connection with avatars.
- Applied **Dependency Injection** and **Reactive Programming** for modular, scalable system design and smooth UX.





Augmented Reality Filters

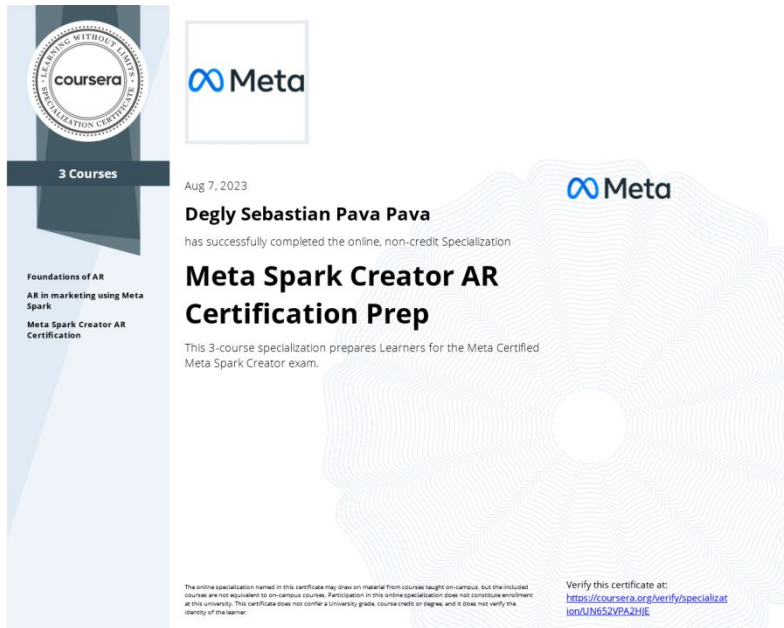
Augmented Reality Filters

- Created AR filters for **TikTok**, **Snapchat**, and **Instagram**, generating over **291 mil lifetime interactions**.
- Designed interactive effects using **Lens Studio**, **Effect House**, and **Spark AR**, tailored for high user engagement.
- Certified as a **Meta AR Developer** and **Spark AR Creator**, demonstrating platform expertise and industry recognition.

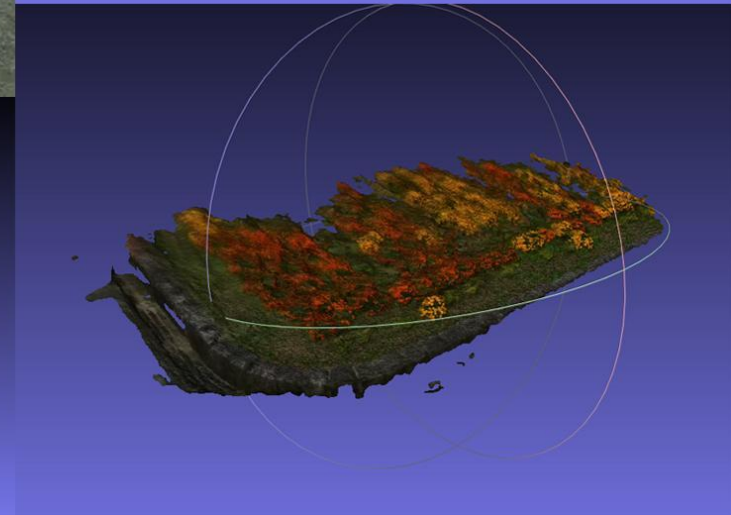
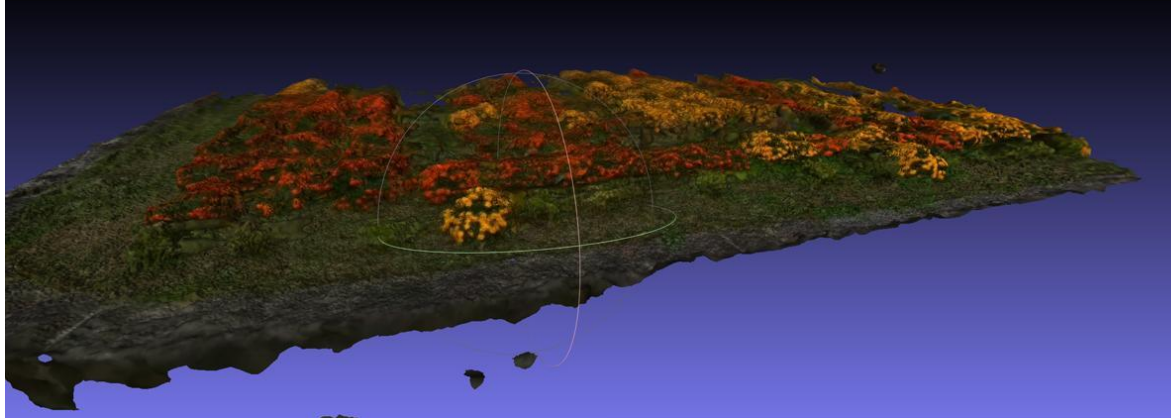
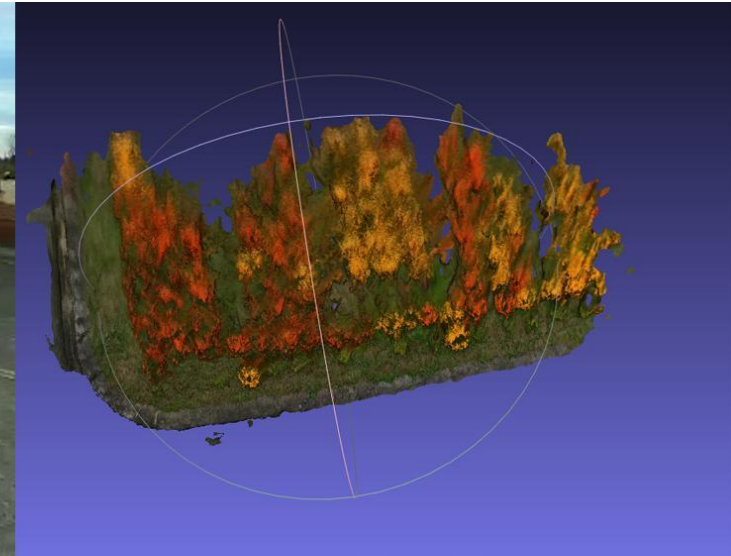


AR Certificates

Obtained through Coursera, more certificates at the end



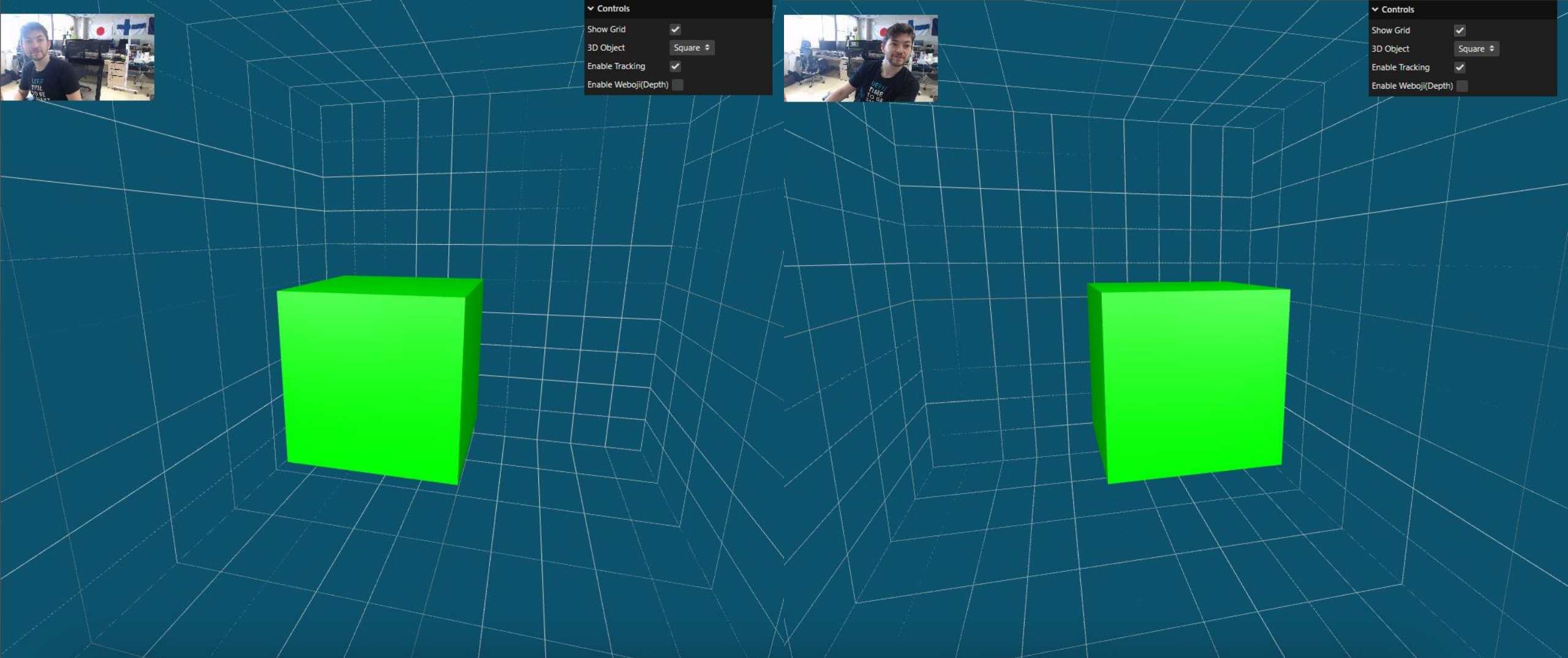
3D RECONSTRUCTIONS USING COLMAP AND MESHLAB



3D RECONSTRUCTIONS USING COLMAP AND MESHLAB

- Reconstructed a full outdoor scene by processing **1,983 video frames** using **COLMAP** for feature extraction and mesh generation.
- Converted data into a usable 3D model via **MeshLab**, then imported into **Blender** for spatial refinement and visualization.
- Demonstrated expertise in **photogrammetry**, **mesh processing**, and **3D reconstruction** for potential use in **AR scanning** of real-world environments and objects.

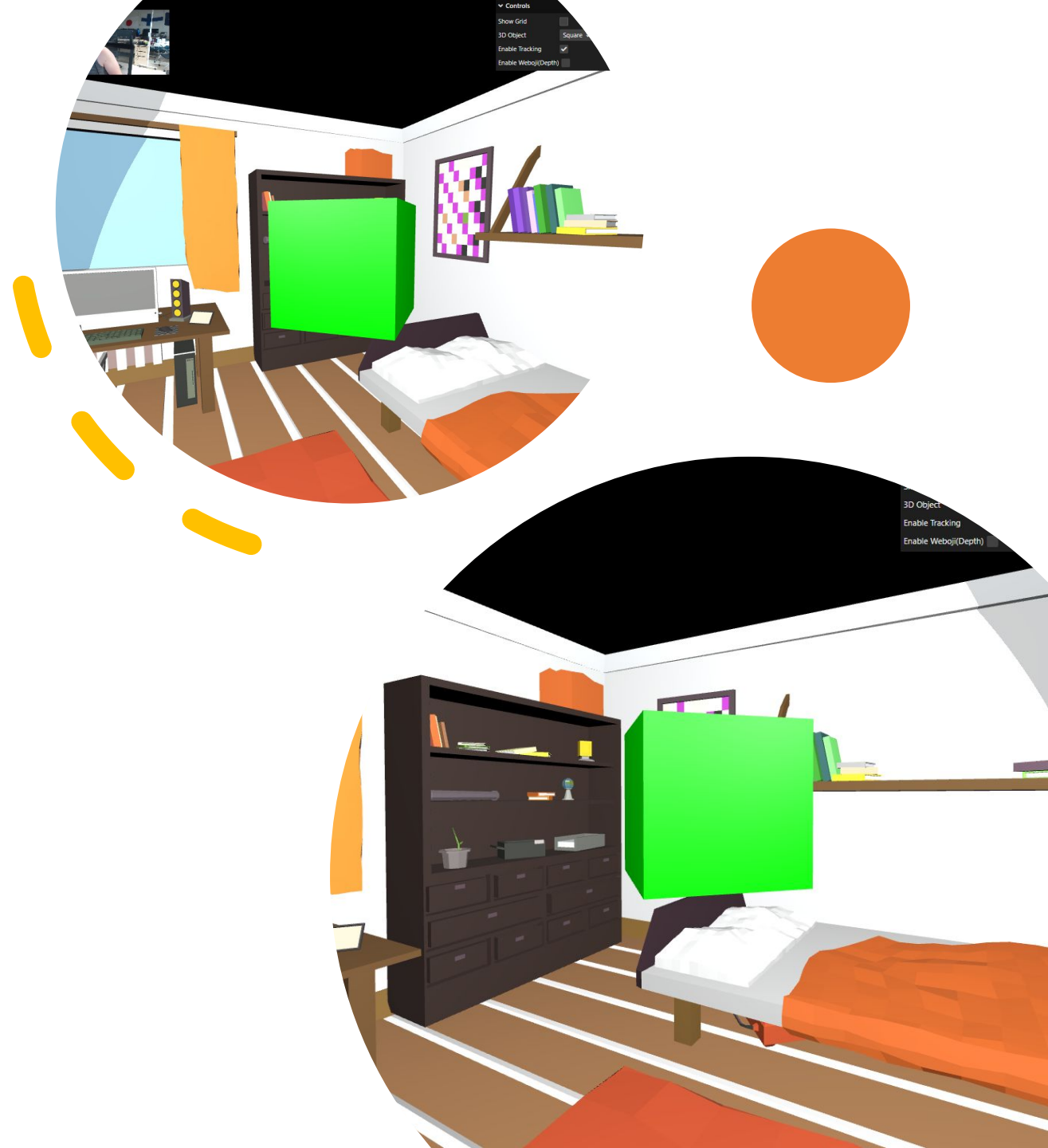




Using MediaPipe to create a
Parallax effect

Using MediaPipe to create a Parallax effect

- Captured and processed **facemesh data** in real time using **MediaPipe** and a standard webcam.
- Tracked subtle head movements to generate a dynamic **Motion Parallax effect**, simulating depth without a 3D display.
- Bridged **computer vision** and **interactive design** to create immersive AR-like visuals using only 2D hardware.



REALTIME RENDERED 3D VISUALIZER



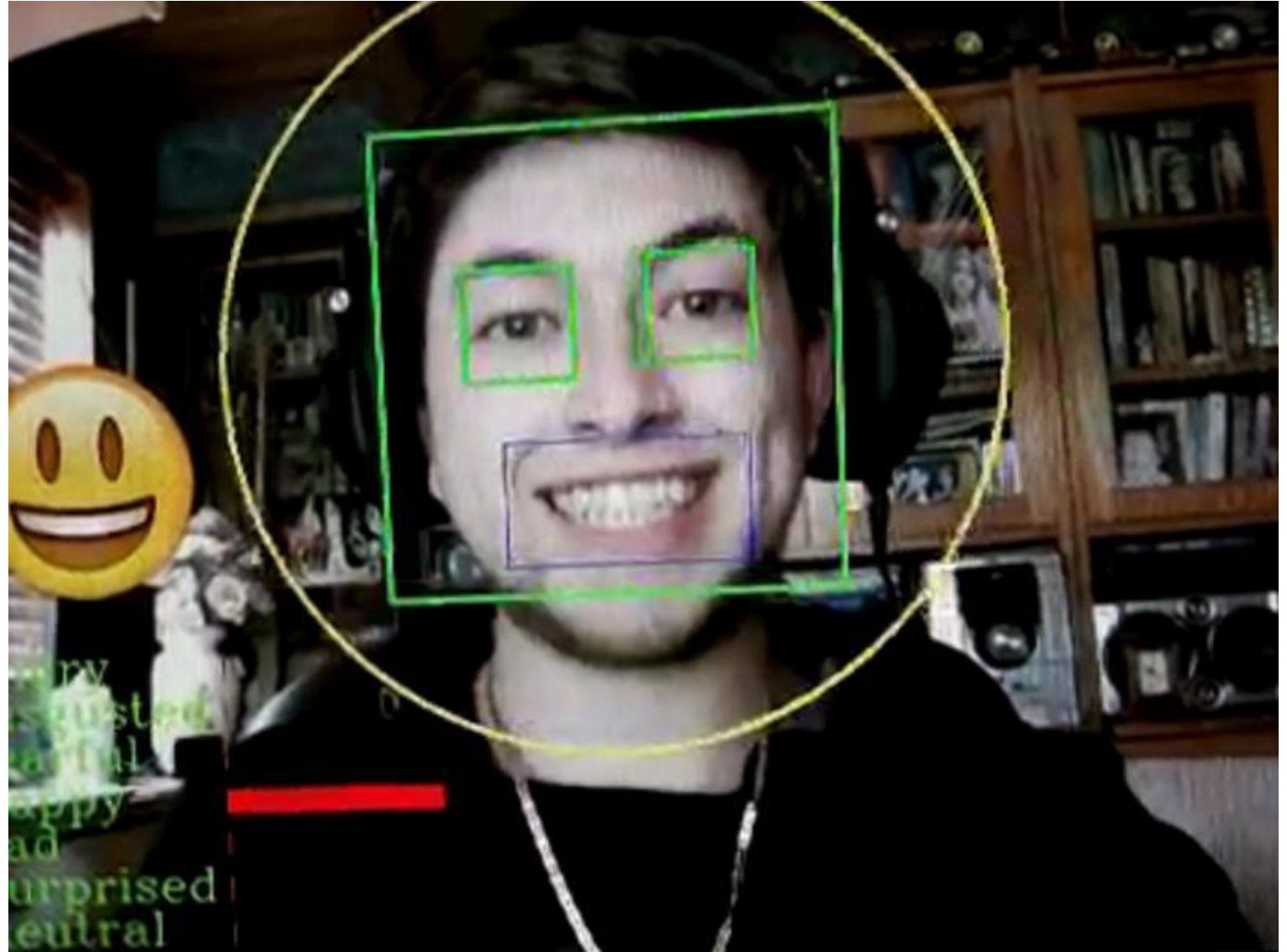
Real-Time 3D Visualizer

(Web + VR)

- Built a real-time 3D visualizer using **Three.js** and **GLSL**, simulating light elevation and color composition from video input.
- Extended the experience to **VR**, with support for **Oculus Quest 2** for fully immersive exploration.
- Showcased expertise in **WebGL**, **shader programming**, and interactive design—bridging web simulation and immersive tech.

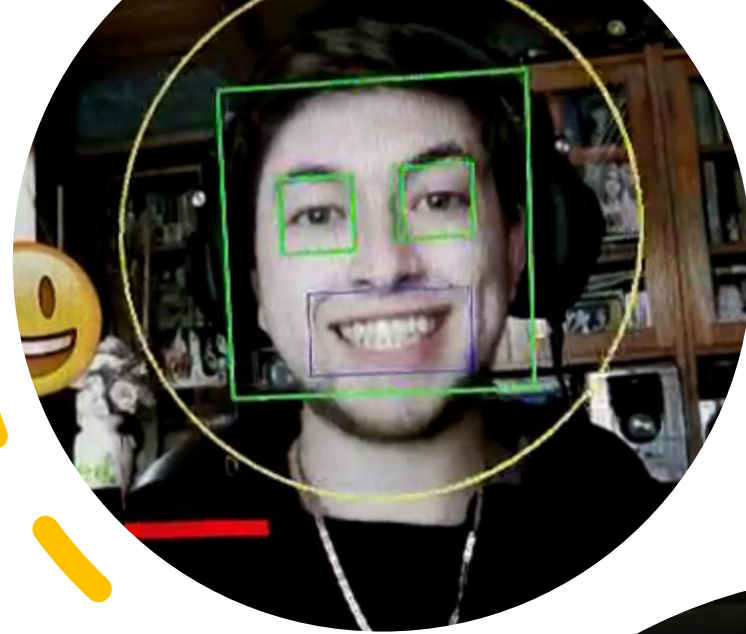


Facial Emotion Recognition with Deep CNN



Facial Emotion Recognition with Deep CNN

- Built a facial emotion recognition system using **Deep Convolutional Neural Networks**, **OpenCV**, and **CUDA**, achieving **65% accuracy**.
- Translated detected emotions into matching **emojis**, creating a playful and intuitive visual output.
- Demonstrates skills in **machine learning**, **computer vision**, and emotional UX design — plus a fun demo video is live on my website!



Certificates

